Face Symmetry Measurement

Understanding the whole project, part by part

A study guide for presenting and defending the project

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How to use this guide

Everyone should read **Part 1** (the one-page summary) and **Part 9** (questions). Each part then stands on its own: read the parts for the slides you present. All numbers are produced by the project's scripts and are identical to those in the paper and slides. Boxes marked *Say it like this* give wording you can use in the presentation; boxes marked *Be careful* list claims you must *not* make.

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1 The project in one page

What it does. You give the system one photo of a face. It finds points on the face (eye corners, nose, mouth, jaw…), compares the left half with the mirror image of the right half, and reports how far the face is from perfectly mirror-symmetric. Before it reports anything, it checks whether the photo is good enough. It answers **accept**, **caution** or **reject**, and explains why.

Input: one RGB photo containing a face.
Output: accept/caution/reject with reasons; a global asymmetry number A (in % of the eye-corner distance); values for brows, eyes, nose, mouth and jaw; a stability band; pictures (overlays); a CSV/JSON export.
Constraints: 2D image only, no model training, runs on a laptop CPU (46 ms per image), and it is *not* a medical, beauty or identity score.

Why it is needed. Existing tools always return a number, even for photos that cannot support one. We found that on ordinary in-the-wild photos the measured asymmetry follows how far the head is turned almost perfectly ($\rho = 0.91$), even when the points are placed by human experts. Without a check, a "symmetry score" of a selfie mostly measures head pose.

Our four contributions.

1. **Correct pairs and a fitted mirror line.** Each point is compared with its true anatomical twin, and the mirror line is estimated rather than assumed vertical. The older nearest-neighbour (KD-tree) matching picks the wrong twin 58.3% of the time.
2. **The mirror test.** We run the detector on the photo *and* on its mirror image. A face's asymmetry cannot change when the photo is flipped, so if the two results disagree, the landmarks are unreliable. This needs no ground truth.
3. **A quality and pose gate** with seven signals, whose thresholds were fixed a priori or learned on training images only.
4. **A way to evaluate without symmetry labels:** compare with the same measurement made from expert-placed landmarks, and degrade photos in 26 controlled ways.

Headline result (LFPW test set)	Value
Agreement with expert-landmark reference, baseline → ours (Spearman ρ)	0.823 → 0.947
Wrong partners chosen by the baseline's KD-tree matching	58.3%
Mirror test vs. true landmark error (Spearman ρ)	0.59
Silent failures on degraded photos, no gate → full system	8.1% → 3.6%
Score drift reduction from mirror averaging	29%
Head yaw vs. asymmetry, even from expert landmarks (Spearman ρ)	0.91

Say it like this

"Existing symmetry tools always give you a number. Ours first checks whether the photo can support a measurement, verifies its own landmarks with a mirror test, and measures with the correct anatomical pairs and a fitted symmetry axis. We evaluated it against expert landmarks and under 26 controlled degradations."

2 Background you need

2.1 Facial landmarks

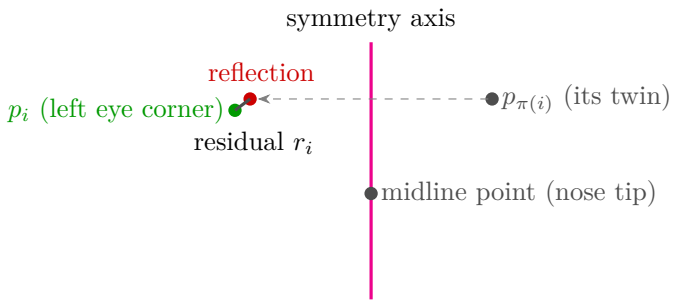
A **landmark** is a named point on the face, such as "outer corner of the left eye" or "tip of the nose". A **landmark detector** is a pretrained model that finds these points in a photo.

- **The 68-point layout** (iBUG / 300-W) is the standard layout of academic benchmarks: 17 jaw points, 10 eyebrow points, 9 nose points, 12 eye points and 20 mouth points. Human experts annotated thousands of photos with it. That is why we report our results in this layout: we can compare against the experts.
- **MediaPipe Face Landmarker** (Google) finds 478 points, a dense mesh covering the whole face (468 face points + 10 iris points). It runs in real time on a CPU and also outputs a 4×4 *transformation matrix* that tells us the head's rotation.
- **The canonical face model** is MediaPipe's "average face" in 3D, with the same 468 vertices. It is *exactly* left–right symmetric. We use this fact to know which vertex is the mirror twin of which (Section 4.2).

2.2 What "symmetric" means geometrically

A face is mirror-symmetric if there is a line (the **symmetry axis** or **midline**) such that reflecting the left half across it lands exactly on the right half.

- **Bilateral pairs**: points that have a twin on the other side (left eye corner ↔ right eye corner).
- **Midline points**: points that are their own twin (nose tip, chin, centre of the lips).
- **Asymmetry**: how far a point is from where the mirror image of its twin says it should be.



The **residual** r_i is the small gap between a point and the reflection of its twin. For a perfectly symmetric face every residual is zero.

2.3 Normalisation: why "% IOD"

Distances in pixels depend on image size: a face photographed close up has bigger gaps. We divide every distance by the **inter-ocular distance (IOD)**, the distance between the two outer eye corners. It is the standard normaliser of the 300-W benchmark. An asymmetry of "2% IOD" means "2% of the distance between the outer eye corners". On an adult face that distance is roughly 9 cm, so 2% IOD is just under 2 mm.

2.4 Head pose: yaw, pitch and roll

- **Roll**: tilting the head towards a shoulder (in-plane rotation). A 2D rotation can undo it, and our fitted axis is unaffected by it.

- **Yaw**: turning the head left or right (shaking "no"). One half of the face gets foreshortened in the photo, and *no 2D operation can undo that.*
- **Pitch**: nodding up or down. It distorts vertical proportions.

This is the core limitation of any 2D symmetry measurement, and the reason the gate checks yaw and pitch.

2.5 Statistics used in the results

- **Spearman** ρ (rank correlation): do two measurements put the faces in the same order? 1 = identical order, 0 = unrelated. It works even when the two measurements use different units, which is why we use it to compare the baseline (face-width units) with ours (IOD units).
- **Pearson** r: linear correlation.
- **MAE** (mean absolute error): the average of $|A_{\text{ours}} - A_{\text{ref}}|$, in % IOD.
- **NME** (normalised mean error): the average distance between detected and expert landmarks, divided by IOD. The standard landmark-accuracy metric.
- **Wilcoxon test**: a paired test of whether one measurement is systematically larger than another.

3 The baseline we improve on (M0)

The closest public tool is `nashokan/FacialAsymmetry` on GitHub. It implements the symmetry part of Varghaei et al. (ICCV Workshops 2025). Since the repository has **no licence**, we did not copy its code; we **re-implemented its method** from its description. Its steps are:

1. Detect the 478 MediaPipe points.
2. Align the face: rotate and scale the image so that vertices 33 and 263 sit at fixed positions, then detect again.
3. Take a vertical midline at the average x of vertices 33 and 263.
4. Reflect every point on the left of the midline, and match it to the *nearest* right-side point (KD-tree nearest-neighbour search).
5. Report the mean matched distance divided by face width.

Its three problems, which we measured:

1. **Wrong partners.** The mesh is dense, so the nearest point after reflection is often a *neighbour* of the true twin. The matching then absorbs part of the asymmetry it should measure. On LFPW, 58.3% of matches are the wrong anatomical twin (Fig. 1).
2. **Assumed vertical midline.** Any tilt of the head, or any error in vertices 33/263, looks like asymmetry. Without the alignment step the baseline's agreement drops from 0.823 to 0.614.
3. **No quality check.** It always returns a number.

(Detail worth knowing: vertices 33 and 263 are the *outer* eye corners, although the baseline's code comments call them inner corners.)

4 Our method, step by step

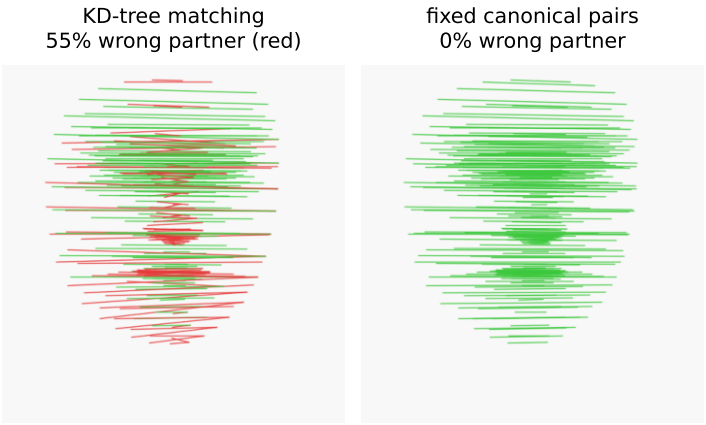


Figure 1: Matches for one face (geometry only). Left: KD-tree matching; red lines are wrong partners. Right: our fixed anatomical pairs.

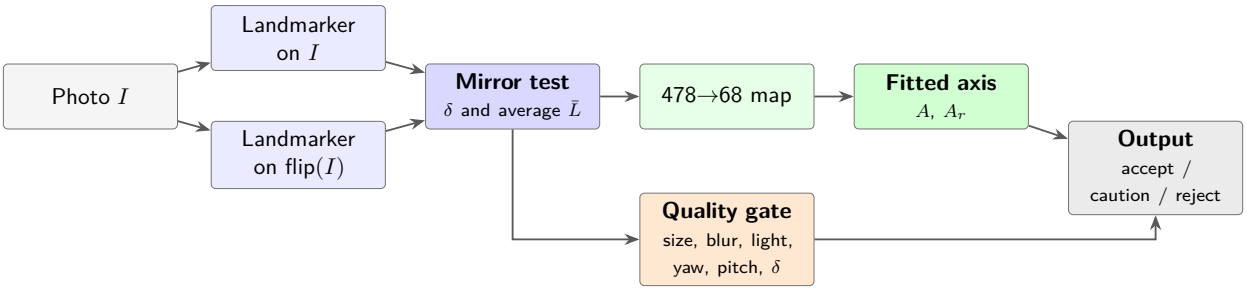


Figure 2: The full system (variant M3). The numbered steps below follow this picture.

4.1 Step 1: detect landmarks (`detector.py`)

What: run MediaPipe Face Landmarker on the photo. We get 478 points in pixel coordinates and the transformation matrix, from which we read yaw, pitch and roll.
Why MediaPipe: it is pretrained (no training needed), fast on a CPU, and its mesh has a *fixed topology*: vertex 33 is always the same anatomical spot, on every face. That is what makes fixed pairs possible.
Limitation: it gives no confidence per landmark (the `visibility` field is empty). The mirror test (Step 5) fills this gap.

4.2 Step 2: find each point's mirror twin (`mesh.py`)

What: take the canonical face model, flip it ($x \to -x$), and for each vertex find the vertex that lands in its place. Because the canonical model is exactly symmetric, this match is perfect (error 0). It gives a **mirror permutation** π: $\pi(i)$ is the twin of vertex i. Result: 220 bilateral pairs and 28 midline vertices (which map to themselves), plus the iris points, paired by hand.
Why: the twin is fixed by anatomy, not guessed per photo. This removes the baseline's wrong-partner problem completely (0% wrong partners).

4.3 Step 3: map the 478-point mesh to the 68-point layout

What: for each of the 68 standard points, choose the mesh vertex that best plays its role. We do this *on training images only*: for every standard point k and every mesh vertex v, we compute how far v typically is from the expert's point k (median over 810 training faces, divided by IOD). Call

this E_{kv}. For a bilateral pair (a, b) of the 68 points we choose

$$v^\star = \arg\min_v \; E_{av} + E_{b\,\pi(v)}, \qquad a \to v^\star, \quad b \to \pi(v^\star).$$

Why this way: choosing the two sides *together*, as mirror twins, guarantees the mapping is itself perfectly symmetric. If the left eye corner were mapped slightly differently from the right eye corner, the mapping alone would create fake asymmetry.
Why 68 points at all: expert annotations exist for these points, so we can check our measurements against experts. The fitted map is also more accurate than the commonly used hand-made map: NME 3.95% vs. 4.23% (inner points), and 8.13% vs. 10.11% on the jaw.

4.4 Step 4: fit the symmetry axis and measure asymmetry (`geometry.py`)

What: instead of assuming the midline is vertical, we search for the line (angle θ, offset ρ) that best explains the face as a mirror image:

$$(\theta^\star, \rho^\star) = \arg\min_{\theta,\rho} \sum_i \left\| \mathcal{R}_{\theta,\rho}(p_{\pi(i)}) - p_i \right\|^2.$$

In words: reflect each point's twin across a candidate line, measure how far it lands from the point, and pick the line that makes these gaps as small as possible overall. We start from a closed-form guess (midpoints and directions of the pairs) and refine it with the Nelder–Mead optimiser. Only the 51 inner-face points are used to fit it.

Then, with residual r_i = reflection of the twin minus the point:

$$a_i = \frac{\|r_i\|}{2 \cdot \mathrm{IOD}}, \qquad A = \text{mean of } a_i \text{ over the 51 inner points}, \qquad A_r = \text{mean over region } r.$$

The factor 2 is there because a gap r_i is shared by two points. a_i is how far point i is from the closest perfectly symmetric face.

Why fit the axis:

- It is **rotation-invariant**: tilt the photo and A does not change (a unit test checks this for $-30°$ to $+25°$, with scaling and shifting).
- It does not depend on two special points being correct.
- It needs no second detection pass, unlike the baseline's align-and-re-detect step.

Worked example (this is one of our unit tests)

Take a perfectly symmetric face, so $A = 0$. Move the inner corner of one eye 2 pixels outward. The fitted axis barely moves, the gap between that corner and its reflected twin becomes about 2 pixels, and each of the two corners gets $a_i \approx 1$ pixel / IOD. The *Eyes* region now has the largest regional value, as it should.

Why the jaw is excluded from A: the 17 jaw points trace the visible outline (silhouette) of the face. Their position along the cheek shifts with head pose, and MediaPipe's outline does not correspond to them point for point (jaw NME 8.13% vs. 3.95% for the inner points). The jaw is still shown as a region.

4.5 Step 5: the mirror test

What: run the detector a second time, on the horizontally flipped photo. Then map the result back:

- a point at x' in the flipped photo is at $W - x'$ in the original (W = image width; MediaPipe uses continuous coordinates, so it is $W - x'$, not $W - 1 - x'$, which we checked: using $W - 1$ left a systematic 1-pixel offset);
- flipping swaps sides, so the vertex the detector calls j in the flipped photo is the original's vertex $\pi(j)$.

This gives a second set of landmarks U for the *same* face. A perfect detector would give $U = L$.

What we get from it:

1. **Mirror inconsistency** δ = mean distance between L and U / IOD. Large δ means the detector is unsure of the geometry. It correlates with true landmark error ($\rho = 0.59$), and it needs no labels.
2. **Averaged landmarks** $\bar{L} = \frac{1}{2}(L + U)$. If the detector systematically pushes points to one side, the flip pushes them the other way, and the average cancels it. This reduces score drift on degraded photos by 29%.
3. **Stability band** $|A(L) - A(U)|$, shown to the user as "±".
4. **Per-region inconsistency** δ_r, used to grey out unreliable regions.

Say it like this

"Flipping a photo cannot change how symmetric a face is. So if the detector gives different geometry for the photo and its mirror image, the detector is unsure, and we can measure that without any ground truth."

Be careful

Flip test-time augmentation already exists in human-pose estimation, where it is used to improve accuracy. Do not claim flipping itself is new. What is new here is using it, for a *symmetry* measurement, as a reliability signal and to cancel left/right bias.

4.6 Step 6: the quality and pose gate (`quality.py`)

The gate looks at seven things the system can actually measure:

Signal	How it is computed	Why it matters
Face size	IOD in pixels	tiny faces: 1 pixel is a large % of IOD
Sharpness	$\log_{10}$ variance of the Laplacian of the face crop at 256 px width	blur moves landmarks
Brightness (low/high)	mean grey level of the face crop	dark or burnt-out faces
Clipped pixels	share of crop pixels near black or white	lost detail
|yaw|	from the transformation matrix	foreshortens one side
|pitch|	from the transformation matrix	distorts proportions
Mirror inconsistency δ	Step 5	detector unsure

Decision rule: each signal has a caution limit and a reject limit; the limits are stored in the file `configs/gate_thresholds.json`. The *worst* signal decides the status, and every triggered signal is listed with a plain-language reason. A rejected photo shows no score unless the user explicitly asks to see it, and it is then labelled unreliable.

How the limits were set (important for the viva):

- *A priori*, from the measurement's assumptions: yaw 10°/20°, pitch 15°/25°, IOD 60/30 px.

- *From the training split*: percentiles of the 810 training faces (2.5th percentile for sharpness and brightness; 90th/99th for δ; 97.5th for region validity).
- The test split was **never** used to choose any limit.

Roll is not gated because the fitted axis already handles it.

4.7 Step 7: the readable index

The app also shows $S = 100/(1 + \alpha A)$ with $\alpha = 0.467$, chosen so that the median training face (A = 2.14% IOD) scores 50. It is only a friendlier way to show A (higher = more symmetric). **It is not a validated clinical or beauty scale.**

4.8 The four variants we compare

M0	Baseline: KD-tree matching, vertical midline, align + re-detect, / face width
M1	Fixed pairs + 68-point map + fitted axis, / IOD (no gate)
M2	M1 + quality/pose gate (without the mirror test)
M3	Full system: mirror-averaged landmarks, gate including δ, region validity

All four are computed from the *same detections* of each image, so any difference comes from the method, not from detector randomness.

5 The data

Dataset: LFPW (Labelled Face Parts in the Wild), part of the 300-W benchmark. Each image has 68 landmarks placed by human experts. We used the copy published on GitHub by the Menpo project (`github.com/menpo/lfpw-train`, `lfpw-test`).

- **Train split** (811 images, 810 faces detected): used *only* to fit the 478→68 map and to set gate limits.
- **Test split** (224 images, all detected): every reported number.
- Each photo is cropped around the annotated face first, as in the 300-W protocol. Many LFPW photos contain several people, and without the crop the detector sometimes analysed the wrong person (this initially caused a 32% mean landmark error, which dropped to 3.95% after the crop).

Why not WFLW? It was the first choice (it has pose, blur and occlusion subsets), but its images are hosted on Google Drive, which our compute environment could not reach. The code reads any 68-point `.pts` dataset, so WFLW or other 300-W subsets can be added later.
Licences and privacy: 300-W is for research use only. No dataset image is stored in our repository or shown in the paper or slides; figures draw landmark geometry only.

6 The experiments and what they show

6.1 The central difficulty: there is no "true symmetry" label

No public dataset says "this face is 2.3% asymmetric". So we built two kinds of test.

1. **Reference from expert landmarks (A_{ref}).** We apply *exactly the same* measurement to the 68 points placed by human experts. If our detector-based measurement agrees with A_{ref}, our pipeline measures what an expert-landmark measurement would. A_{ref} is not perfect truth (experts make small errors too), but it is the best reference available.
2. **Controlled perturbations.** We degrade each test photo in known ways (brightness, blur, covering a part, rotation, shrinking). The face itself does not change, so the correct answer does not change either. Any change in the score is an error.

Two metrics we defined (know these well):

- **Silent failure**: the system *reports* a score (does not reject), but it is off from A_{ref} by more than 1% IOD. This threshold was fixed before running and is about twice the normal error. This is the failure a gate exists to prevent.
- **Drift**: the median change of the score between the clean and the degraded photo, divided by how much scores normally vary between people (the SD of clean scores). Dividing by the SD lets us compare the baseline (face-width units) with ours (IOD units).

6.2 E1–E3: do we agree with the experts?

Measure (n=224)	ρ	r	MAE
M0 baseline (KD-tree, vert. midline)	0.823	0.762	–
M0 w/o alignment	0.614	0.630	–
M0 + fixed pairs (no KD-tree)	0.931	0.903	–
M1: 68-pt pairs + fitted axis	0.938	0.951	0.45
M3: M1 + mirror averaging	0.947	0.957	0.45

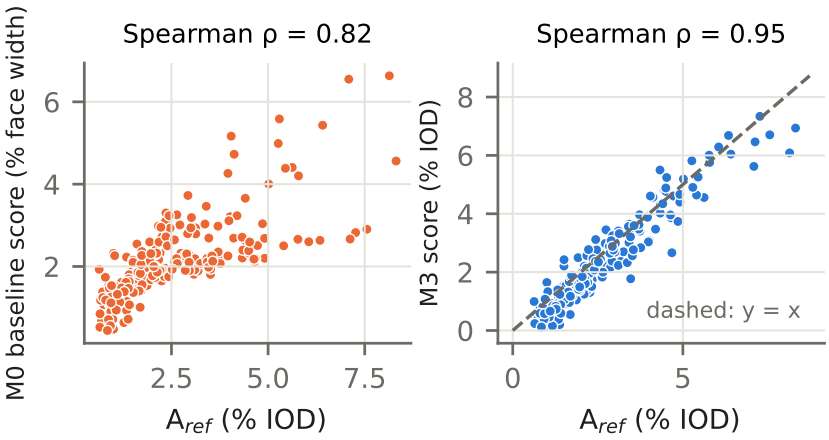


Figure 3: Each dot is one test face. Left: baseline. Right: ours, close to the $y = x$ line.

How to read it: replacing *only* the KD-tree matching with fixed pairs (same detections, same midline) already raises ρ from 0.823 to 0.931. So *wrong correspondence was the baseline's main problem.* The fitted axis on the 68-point layout gives 0.938 (average error 0.45% IOD), and mirror averaging gives 0.947.

Regions and shrinkage (E3). Regional agreement is 0.84–0.98. The detector's asymmetry is systematically *smaller* than the experts': by 12.4% overall and 26.7% at the mouth (statistically significant everywhere except the brows). MediaPipe has a built-in "average face" prior that pulls unusual (asymmetric) shapes towards symmetry. So our regional numbers should be read as conservative.

6.3 E8: head pose dominates in-the-wild asymmetry

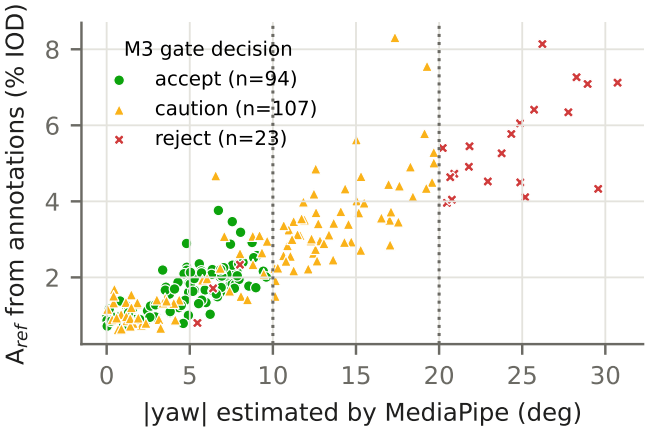


Figure 4: Asymmetry from expert landmarks against head yaw. Dotted lines: gate limits.

|yaw| (deg)	n	A_{ref}	M1	|err| M3	NME
0-5	77	1.20	0.80	0.47	3.03
5-10	67	2.14	1.86	0.40	5.51
10-20	60	3.60	3.41	0.42	3.32
$\geq$20	20	5.50	5.42	0.57	4.13

How to read it: even from expert landmarks, asymmetry rises from 1.20% to 5.50% as yaw grows ($\rho = 0.91$). This is a property of 2D photos, not of our detector. The detector's error ("|err| M3") stays about the same at every yaw. That is exactly why a separate pose check is needed: a check that only looks at detector error would not catch it. Among gate-accepted faces the correlation falls to 0.75.

Be careful

The pose groups contain *different people* (each LFPW person appears once). Do not say "the same face measured at different angles".

6.4 Does the mirror test find bad landmarks?

Risk–coverage curve: sort images from most to least trusted by a signal and keep the top x%. A good signal keeps low-error images first. The area under the curve (AURC) summarises it: lower is better.

- For landmark error (left): random 3.37, gate signals without the mirror test 3.29, **mirror test 2.99**, perfect oracle 2.75. The mirror test closes 61% of the gap between random and perfect.
- For the asymmetry error on clean photos (right), no signal does much better than random (0.418 vs. 0.442). On clean photos, most landmark errors are similar on both sides and cancel in A. The gate pays off on degraded photos (next section).

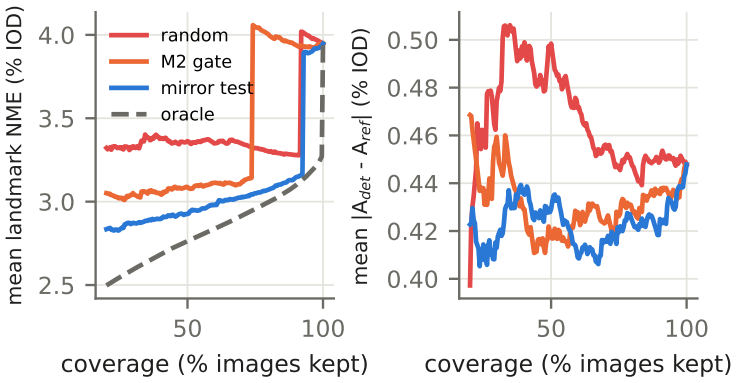


Figure 5: Risk–coverage. Keep the images the signal trusts most; the y-axis is the average error of the kept images (lower is better).

6.5 E4–E7: robustness to degraded photos

26 settings × 224 faces = 5,824 degraded images: brightness ×0.25 to ×2.5, blur of 1–8% of IOD, a grey box over one eye/nose/mouth, rotation ±5–30°, and shrinking to 50–12% of the size. **How to**

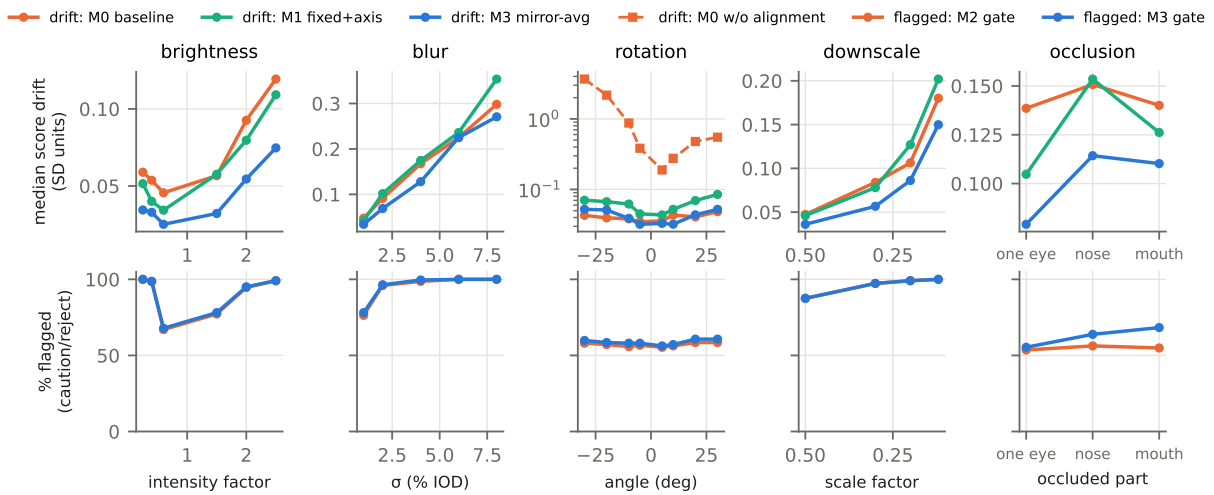


Figure 6: Top: drift (lower is better). Bottom: % of images the gate flags.

read it:

- MediaPipe still finds a face in at least 79.0% of images in every setting. So the risk is not "no answer" but a *plausible-looking wrong answer*.
- Blue (mirror averaging) drifts least under every type of degradation.
- Rotation panel (log scale): the vertical-midline method without alignment drifts by up to 3.67 SD. Our fitted axis stays below 0.052 SD with no re-detection.
- The gate flags more and more images as blur, darkness or shrinking increase, which is what we want.

6.6 E9: ablation (what each component contributes)

Setting	Var.	n	Reported (%)	Accepted (%)	ρ vs ref	Silent fail. (%) ↓	Drift (SD) ↓
Clean test	M0	224	100.0	100.0	0.823	–	–
	M1		100.0	100.0	0.938	5.8	–
	M2		90.2	43.3	0.921	4.9	–
	M3		89.7	42.0	0.932	4.0	–
Perturbed, pooled	M0	5824	98.4	98.4	0.814	–	0.070
	M1		98.4	98.4	0.921	8.1	0.076
	M2		64.0	22.6	0.918	3.4	=
	M3		63.2	21.0	0.927	3.6	0.054
blur	M0	1120	94.0	94.0	0.787	–	0.131
	M1		94.0	94.0	0.870	14.4	0.130
	M2		37.0	5.8	0.903	2.7	=
	M3		35.6	5.2	0.919	2.3	0.099
downscale	M0	896	98.7	98.7	0.821	–	0.087
	M1		98.7	98.7	0.909	9.2	0.097
	M2		33.5	4.0	0.919	1.8	=
	M3		33.4	4.0	0.927	1.9	0.066
brightness	M0	1344	99.5	99.5	0.818	–	0.065
	M1		99.5	99.5	0.936	5.8	0.056
	M2		62.8	10.6	0.927	2.8	=
	M3		62.4	10.2	0.940	2.7	0.037
occlusion	M0	672	98.8	98.8	0.804	–	0.142
	M1		98.8	98.8	0.919	8.8	0.124
	M2		87.9	45.1	0.900	6.2	=
	M3		85.1	37.5	0.911	6.0	0.098
rotation	M0	1792	100.0	100.0	0.825	–	0.039
	M1		100.0	100.0	0.944	5.2	0.059
	M2		88.1	43.0	0.924	4.0	=
	M3		87.7	41.4	0.928	4.9	0.040

How to read it: without a gate (M1), 8.1% of degraded measurements are silent failures (14.4% under blur). With the gate (M2/M3) this falls to 3.4%/3.6% (2.3% under blur). The price is coverage: only 63.2% of degraded and 89.7% of clean images still get a score. M3 has the lowest drift (0.054 vs. 0.076 SD).

6.7 Gate behaviour on clean photos, and an honest negative result

On the clean test set M3 accepts 42.0%, cautions 47.8% and rejects 10.3% of faces. Pose is the main trigger: LFPW faces are mostly turned. Accepted faces have a lower landmark error (3.03% vs. 4.71% NME).

Negative result: when a grey box covers the nose or mouth, the region is flagged unreliable in 17.5% and 18.4% of faces. When it covers an *eye*, it is flagged in only 0.9%. MediaPipe "invents" the hidden eye from its average-face prior, the same way in both the photo and its mirror, so the mirror test sees agreement. Lesson: *the mirror test catches inconsistent landmarks, not confidently invented ones.* An appearance-based occlusion check is future work.

7 The code, file by file

File	What it does
`src/fsm/detector.py`	Loads MediaPipe once; `detect(img)` returns 478 points, the pose matrix and the number of faces; `pose_from_transform` turns the matrix into yaw/pitch/roll.
`src/fsm/mesh.py`	Mirror permutation from the canonical model; the 68-point pairs and regions; the fitted 478→68 map.
`src/fsm/geometry.py`	All the maths: reflection, fitted axis, asymmetry a_i, A, A_r; the baseline KD-tree method; un-flipping and mirror inconsistency.
`src/fsm/quality.py`	Sharpness/brightness/clipping on the face crop; `gate_decision` (accept/caution/reject + reasons); `risk_score`.
`src/fsm/pipeline.py`	`raw_detect` (all detector calls) and `measure_from_raw` (all geometry); `variant_outputs` turns one measurement into M0–M3.
`src/fsm/viz.py`	Drawing: mesh, axis, pairs, residuals, heatmap, status banner.
`src/fsm/data.py`	Reads LFPW `.pts` files; crops to the annotated face.
`app.py`	Streamlit interface; it only *calls* the engine, with no maths in the UI.
`experiments/`	Run in order: `cache_detections` → `fit_mp68_mapping` → `calibrate_gate` → `run_clean_eval` → `run_perturbations` → `summarize_perturbations` → `make_figures`, `make_tables`.
`configs/`	Canonical face model, fitted map, gate limits (JSON); no thresholds are hard-coded in the UI.
`tests/test_geometry.py`	15 unit tests: reflection, zero score for a symmetric face, rotation/scale invariance, known displacement, un-flip round trip, mirror-consistent map, gate levels.

Design choices you may be asked about: the engine never imports Streamlit, so the same code runs the app and the experiments. Detections are cached once, so every variant is scored on identical landmarks. Every table and figure in the paper is written by a script from saved CSVs, so no number was typed by hand.

8 The app, screen by screen

- **Sidebar**: choose upload or webcam; "stress-test" sliders (brightness, blur, rotation, occlusion) to show the gate reacting live; a toggle to show the baseline's number.
- **Verdict**: green/amber/red box with the reasons (for example "head is turned left/right"). For reject the score shows "withheld".
- **Numbers**: asymmetry A (% IOD), mirror stability ±, symmetry index (0–100).
- **Overlays tab**: mesh; fitted axis with pairs; green point vs. red reflection of its twin; heatmap (grey = unreliable region).
- **Regions tab**: table and bar chart for the five regions, with reliable/unreliable status.
- **Quality signals tab**: each signal's value next to its caution/reject limits, and the pose angles.
- **Export tab**: JSON and CSV download.

9 How to present it

The story in four sentences: (1) Symmetry tools give a number for every photo, but 2D photos are easily dominated by pose and image quality. (2) We measure with correct anatomical pairs and

a fitted axis, check the landmarks with a mirror test, and gate on quality and pose. (3) Against expert landmarks and under 26 degradations, this agrees much better and prevents most silent failures. (4) We are honest about what it cannot do: occluded eyes, shrinkage, 2D pose.

Suggested split (change to match who actually did what): slides 1–5 Pankaj; 6–9 Prutha and Ashish; 10–14 Ankit and Ashish; 15 Pankaj; 16 (demo) Mahi; 17 Prutha. The speaker notes on every slide hold the details, and `presentation/DEMO_SCRIPT.md` has the step-by-step demo.

Timing for about 12 minutes: problem and motivation 2 min; related work and contributions 1.5 min; method 3 min; results 3 min; failures 1 min; demo 2–3 min.

10 Questions you should be ready for

What exactly is novel?
Not landmarks, regions or the UI; those exist. Our novelty is (1) fixed anatomical pairs plus a fitted axis, which fixes the 58.3% wrong-partner problem and raises agreement from 0.823 to 0.947; (2) the mirror test as a label-free reliability signal and bias-cancelling average; (3) a train-calibrated gate evaluated by silent failures and coverage; (4) the evaluation protocol.

How do you know your score is right without labels?
We compare with the identical measure on expert landmarks, and we use degradations where the answer is known not to change. We do not claim agreement with human symmetry *ratings*; we have none.

Why not just align the face like the baseline?
Alignment needs a second detection and still relies on two points being correct. The fitted axis is mathematically rotation-invariant and uses all 51 inner points.

Why IOD and not face width?
IOD is the 300-W standard and does not depend on the jaw outline, which moves with pose.

Did you tune thresholds on the test set?
No. Pose and size limits were set a priori; the rest are training-set percentiles. The test set was only evaluated.

Why does the gate reject so many LFPW faces?
LFPW is in-the-wild: about two thirds of test faces are turned more than 5°. Given $\rho = 0.91$ between yaw and asymmetry, refusing those is correct behaviour.

Why not train a deep network?
There are no symmetry labels to train on, and explicit geometry is interpretable and unit-testable. Training a detector was out of scope.

Could MediaPipe have seen LFPW in training?
Possibly; its training data is not public. That is why we do not present its NME as an independent benchmark result, only as a comparison between our mappings and signals.

What is the 0–100 index?
A monotone re-expression of A for readability, with the median training face at 50. It is not a validated scale.

Is this a medical or attractiveness score?
No. It is a 2D geometric indicator, and the app says so.

Did you copy code?
No. The audited repositories have no licence, so we re-implemented the baseline method ourselves. The MediaPipe model and canonical mesh are Apache-2.0.

What would you do next?
Use MediaPipe's 3D landmarks to undo yaw and validate on AFLW2000-3D (which has true pose); add an appearance-based occlusion check; use WFLW subsets; collect consented multi-pose photos with human symmetry ratings.

11 Glossary

A / A_r global / regional asymmetry, in % IOD.

A_{ref} the same measure computed from expert landmarks (the reference).

AURC area under the risk–coverage curve; lower is better.

Bilateral pair a point and its mirror twin on the other side of the face.

Canonical face model MediaPipe's exactly symmetric 3D average face.

Caution / reject the gate's warning level / refusal level.

δ, δ_r mirror inconsistency (whole inner face / one region), as a fraction of IOD.

Drift median score change under degradation, in units of the between-person SD.

IOD inter-ocular distance (outer eye corner to outer eye corner).

KD-tree a data structure for fast nearest-neighbour search; the baseline uses it to match points.

LFPW / 300-W the benchmark with 68 expert landmarks per face that we used.

M0–M3 baseline, fixed pairs + axis, + gate, full system.

NME landmark error divided by IOD.

π the mirror permutation: $\pi(i)$ is the twin of point i.

Silent failure a reported score that is off by more than 1% IOD.

Yaw / pitch / roll turning left–right / nodding / tilting to a shoulder.